

Impacts of climatic stability on the structural and functional aspects of macroinvertebrate communities after severe drought

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SUMMARY

1. Unpredictable, extreme climatic events (e.g. droughts) can potentially destabilize aquatic communities. From 1998 to 2002, southwestern Georgia, U.S.A., experienced the third worst drought of the last 100 years, leading to loss of surface flow in many small streams. We sampled macroinvertebrates, flow and water chemistry in small headwater streams from 2001 to 2007 in two adjacent coastal plain streams of contrasting headwater type (wetland and groundwater-fed seep) following resumption of flow.
2. Decreasing water temperature, conductivity and nutrient concentrations within the first 2 years of the study indicated flushing of the streambed associated with increased discharge. Invertebrate community composition became less variable over time and during wetter years, reflecting water chemistry, hydrological and climate conditions.
3. A core set of species appeared immediately following breaking of the drought in both streams, reflecting a shared species pool. These species exhibited resilience traits, including short life cycles and resistance to desiccation, which allowed for rapid recovery from disturbance. Such species, which were small-bodied, sclerotized and abundant in the drift, were then replaced as flows increased by those that were larger, soft-bodied and rare in drift, suggesting a more stable and less ephemeral habitat.
4. Hydrologic regime and long-term precipitation indices were strongly correlated with invertebrate community and trait structure. Long-term data allowed for better interpretation of the effects of infrequent disturbances on aquatic ecosystems. Additionally, long-term precipitation indices (i.e. 48-month standardized precipitation index) can indicate the likelihood of a return to drought, allowing for the collection of pre-disturbance data.

Keywords: disturbance, drought, headwater streams, macroinvertebrates, traits

Introduction

Natural disturbances affect community structure and ecosystem function and partly shape aquatic and

terrestrial communities (Sousa, 1984; Resh *et al.*, 1988). Aquatic ecosystems are especially vulnerable to extreme climatic changes, such as drought, because these disturbances alter the flow regime, water chemistry and, ultimately, the biotic community (Wood & Petts, 1999). The long-term effects of drought on the economy, wildlife habitat and recreation occur as ramp disturbances, sometimes over periods of years (Lake, 2003), as opposed to the effects of flooding

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events, which subside after weeks or months. The frequency and predictability of droughts are generally low. However, when drought does occur, it can potentially destabilize aquatic communities. The forecast for climate change suggests an increased frequency of extreme events, particularly drought, over the next century, especially in Asia, southern Australia and Europe, and the southern and western regions of the U.S.A. (Wetherald & Manabe, 2002; Kundzewicz *et al.*, 2007). Increased intensity and frequency of natural disturbances will ultimately affect ecosystem stability and influence the resistance and resilience of communities to change.

During extreme drought, streams typically form a series of disconnected pools and surface flow ceases, a response that can potentially reset the aquatic community. Furthermore, accumulation of nutrients and toxic waste (Towns, 1985, 1991; Closs & Lake, 1995; Dahm *et al.*, 2003), coupled with increased temperature (Matthews, 1998) and lowered dissolved oxygen (Stanley, Fisher & Grimm, 1997; Golladay & Battle, 2002), stress the remaining species. Survival after drought depends on specific life-history traits, including resistance to desiccation and the ability to colonize habitats rapidly through drift, aerial migration and/or oviposition (Williams, 1987, 1996; Boulton, 1989). Further, colonization reflects subsequent changes in water chemistry, habitat availability and food resources following flow resumption.

Biological traits may be better indicators of ecosystem function than are changes in abundance of individual species and they are expected to change across a gradient of anthropogenic and natural disturbances (Dolédec, Statzner & Bournard, 1999; Charvet *et al.*, 2000; Statzner, Hildrew & Resh, 2001). However, species loss decreases the ability of ecosystems to resist disturbance and leads to lowered stability (Hooper *et al.*, 2005). Therefore, an integrative approach should use both species composition and biological traits to predict community responses to disturbances (Richards *et al.*, 1997). Specific biological traits represented within a drainage basin are first determined by environmental filters (e.g. climate and geology) creating a template for species and their traits (Townsend & Hildrew, 1994; Poff, 1997). Thus, a subset of traits is expected to respond to disturbances within a certain region. For example, species that are resilient to disturbance display a series of traits, including small size and multiple generations per

year, that promote fast population growth (Townsend & Hildrew, 1994).

We used a 6-year (2001–2007) dataset of macroinvertebrates from headwater streams after an intense drought in the southeastern U.S.A. (1998–2002) to characterize inter-year successional patterns following flow restoration, and in relation to water quality and climatic parameters, biological traits and taxonomic composition and community persistence. Biological traits were expected to respond similarly in the two streams because they are adjacent headwater streams in the same basin and accessible to the same species pool. Additionally, traits were anticipated to respond primarily to local environmental variation (e.g. water quality parameters) as a reflection of large-scale environmental filters. However, changes in regional climatic data were expected to structure the overall successional pattern of the community.

Methods

Site description

The two study streams were located in southwestern Georgia, U.S.A. (30°49'N : 84°37'W), approximately 16 km south of Bainbridge in the Coastal Plain physiographic province. They lie within the Dry Creek catchment, which discharges to the Flint River approximately 22 km upstream of the Jim Woodruff Dam of Lake Seminole. Surface water flow in this basin is lowest from September to November and peaks during January–April, due to higher rainfall and decreased evapotranspiration (Couch, Hopkins & Hardy, 1996). Streams and rivers in the Coastal Plain receive substantial amounts of groundwater because they are typically deeply incised into underlying aquifers (Couch *et al.*, 1996). These streams were first order (width *c.* 1.25 m), perennial, groundwater-influenced, low-to-medium gradient, with a sand-dominated substratum [D_{50} was 0.54 mm in the wetland-fed stream (WF) and 0.71 mm in the seep-fed stream (SF)]. The WF had a broader, flatter valley floor with several lateral wetlands and drained a catchment of 26.2 ha with a gradient of 1.96%. The groundwater-fed seep (SF) was more incised with a steeper, V-shaped valley, a 43.9 ha drainage basin and a 2.11% gradient (Summer *et al.*, 2003). Both catchments are forested with WF dominated by *Nyssa biflora* (Walter), *Liriodendron tulipifera* (Linnaeus),

Pinus taeda (Linnaeus) and *Quercus alba* (Linnaeus) and SF dominated by *Pinus glabra* (Walter), *Fagus grandifolia* (Ehrhart), *L. tulipifera* and *Quercus nigra* (Linnaeus).

Hydrological and environmental variables

The climate of the region is characterized by warm, humid summers and mild winters. Mean temperature in January, the coldest month of the year, ranges from 2.8 to 16.3 °C. July is the hottest month, with average temperatures ranging from 21.5 °C to 33.5 °C (SERCC, 2007). Mean annual precipitation is 1412 mm, with June having the highest mean rainfall (152.1 mm) and October the lowest (77.5 mm) (SERCC, 2004). Summer rains are usually short, with high-intensity events giving way to low-intensity frontal events from late autumn to early spring. Due to close proximity to the Gulf of Mexico, heavy rainfall associated with hurricanes and tropical storms is not unusual in late summer.

Drought characteristics were based on regional precipitation data and flow data from both study streams. Flow data were obtained from in-stream parshall flumes and ISCO (Teledyne Isco, Lincoln, NE, U.S.A.) samplers (Summer *et al.*, 2003), beginning in 2001. A standardized precipitation index (SPI) (McKee, Doeskin & Kieist, 1995) was calculated to assess the frequency and duration of droughts in the region based on monthly precipitation averages for southwest Georgia (National Climatic Data Center, 2007). This index is preferable over the Palmer drought severity index because it is easier to interpret, more realistic over the long-term and does not depend on a normal distribution of precipitation (Guttman, 1999). SPI values <1 indicate a water deficit, while those above 1 an excess. SPI values were calculated based on 3-, 12- and 48-month averages to determine the presence of short-term, intermediate and long-term droughts respectively. For example, the 3-month index for November 2002 is the average of August, September and October 2002.

Water temperature was measured from October 2001 until February 2007 with an Onset HOB0[®] temperature logger (Pocasset, MA, U.S.A.), programmed to record temperature every 15 min. Water chemistry and meteorological measurements have been collected by other investigators as part of the Dry Creek Study and these data were available for use

here. Monthly *in situ* measurements for dissolved oxygen, specific conductance, temperature, pH and turbidity were made at eight sites (two per stream) with portable metres. Grab samples were taken from a midstream location and analysed for inorganic nitrogen, inorganic phosphorus and ammonium. Specific details of data collection and sample analysis can be found in Jones *et al.* (2003). Values were $\ln(x + 1)$ -transformed prior to analysis to normalize data.

Invertebrate sampling

Benthic macroinvertebrates were collected from four sample reaches (two per stream, separated by c. 50 m) with a 500- μ m-mesh D-frame net (0.3-m wide) in December and February for 6 consecutive years beginning December 2001, which marked the return of flowing water in both streams. Twenty dipnet sweeps (c. 0.5 m per sweep) were taken from each reach for a total of c. 3.1 m² area sampled from all available habitats and were combined into a single sample in accordance with the U.S. EPA's macroinvertebrate sampling protocols (Barbour *et al.*, 1999). Samples were preserved in 95% ethanol and identified to genus using regional and national keys. Chironomid larvae were quantitatively subsampled, mounted and identified following Epler (1995) and Merritt & Cummins (1996).

Biological traits

Nine biological traits were selected to characterize body morphology (i.e. size, body shape, body armouring), life history (i.e. voltinism, resistance to desiccation), mobility (i.e. occurrence in drift) and ecology (i.e. rheophily, habits, feeding preferences) (Table 1). These were anticipated to vary in response to changes in precipitation and display low statistical and phylogenetic dependence among them (Poff *et al.*, 2006). Some desired traits were omitted due to the lack of available information (e.g. on fecundity), particularly within the chironomid genera. The nine biological traits were divided into 30 categories (ranging from two to six categories per trait). Trait information was collected from the literature (e.g. Viera *et al.*, 2006), as well as through communication with taxonomic experts. Trait information was coded at the generic level, except for some Diptera and non-insect taxa, which were coded at the family or order respectively.

Table 1 Definition and codes for biological traits and categories

Trait	Code	Category	Trait	Code	Category
Voltinism	v1	Semivoltine	Habit	h1	Clingers
	v2	Univoltine		h2	Burrowers
	v3	Multivoltine		h3	Swimmer
Drying resistance	d1	Absent		h4	Sprawler
	d2	Present		h5	Skater
Drift	df1	Rare		h6	Climber
	df2	Common	Trophic	tr1	Gatherer
	df3	Abundant		tr2	Filterer
Armouring	ar1	Soft		tr3	Scraper/herbivore
	ar2	Sclerotized		tr4	Shredder
	ar3	Case/Shell		tr5	Predator
Maximum size	s1	Small (<9 mm)	Shape	sh1	Streamlined
	s2	Medium (9–16 mm)		sh2	Not streamlined (Tubular)
	s3	Large (>16 mm)			
Rheophily	r1	Standing			
	r2	Slow			
	r3	Fast			

Where information on a particular trait could not be obtained for a taxon (in <5% of cases), zero scores were entered for each category so it did not influence overall results (Chevenet, Dolédec & Chessel, 1994). Individual taxa were then scored for the extent to which they displayed the categories of these traits using a 'fuzzy coding' procedure (Chevenet *et al.*, 1994). Fuzzy coding allows taxa to exhibit trait categories to different degrees (Chevenet *et al.*, 1994) to take account of intraspecific variations in trait expression (Charvet *et al.*, 2000). The scoring range 0–3 was adopted, with 0 being no affinity to a trait category and 3 being high affinity. Traits were rescaled as proportions (sum = 1), such that for each trait category, values ranged from 0 (no affinity among individuals for the category) to 1 (all individuals had exclusive affinity for the category) and categories summed to 1 for each trait. To describe the functional composition of communities in terms of density of individuals, the proportion of each category per trait was multiplied by invertebrate abundance. This resulted in a trait-by-site array that contained the density of individuals for each trait category for each site and was transformed [$\ln(x + 1)$] to approximate a normal distribution for statistical analyses.

Statistical analysis

Environmental variables Environmental variables were analysed over time with repeated measures ANOVA (SAS Institute Inc., 2002). When differences were

significant, *post hoc* analysis was conducted using Tukey's test and Bonferroni corrections.

Persistence: species composition and traits Compositional persistence of invertebrate communities was examined separately for the two streams between pairs of successive years. Persistence was measured by calculating Bray–Curtis distances (Bray & Curtis, 1957) between successive years (i.e. 2004 and 2005; 2005 and 2006) based on abundance data and biological traits. Bray–Curtis distances are a measure of community dissimilarity, with values ranging from 0 to 1. Zero denotes identical samples; thus, higher values denote lower compositional persistence. ANOVA was used to examine between-year differences in compositional and trait persistence scores for the streams. The relationship between Bray–Curtis values and flow and SPI values were regressed to assess the impact of hydrological and climate variables on community and trait persistence.

Ordination: species composition and traits Non-metric multidimensional scaling (NMDS; Kruskal, 1964) was used to explore temporal patterns in species composition and biological traits. NMDS is an ordination method based on ranked distances between samples and is highly suitable for ecological data that typically include zero values. First, a distance matrix was constructed using Sorensen's metrics (i.e. Bray–Curtis), which is useful for heterogenous data and does not place emphasis on outliers. This metric is

proportional-based and defined as the distance between taxa i and h :

$$D_{ih} = \frac{\sum a_{ij} - a_{hj}}{\sum a_{ij} + a_{hj}},$$

where, within a rectangular matrix of taxa and samples, a_{ij} is an element representing the abundance of taxon i in sample j , and a_{hj} is a different element in the matrix, representing the abundance of taxon h within the sample. To reduce the chance of local optima (Legendre & Legendre, 1998; McCune, Grace & Urban, 2002), an initial ordination with 1000 runs was conducted and the ordination with the lowest stress value was used as the starting configuration for NMDS. The number of dimensions retained was evaluated after inspecting the stress (goodness of fit) of solutions with dimensions 1–6, with values close to 0 being a good fit of the data. Significance was assessed by conducting Monte Carlo tests using 999 runs of randomized data in the final ordination. A P -value was calculated as a function of the number of randomized runs that resulted in a stress less than or equal to the observed stress (McCune & Mefford, 1999). Ordinations were performed separately for each stream because preliminary analysis indicated that differences between sites masked any temporal effects. Ordinations were performed on species abundances and abundance-weighted biological traits individually.

A multiresponse permutation procedure (MRPP; McCune *et al.*, 2002) was used to test for significant differences in taxonomic composition and biological trait structure over time at each stream. MRPP is a nonparametric method that examines the null hypothesis of no difference between two or more *a priori* defined groups. The test statistic A describes the degree of within-group homogeneity compared with that expected by chance. MRPP was based on $\ln(x + 1)$ -transformed abundance data and the Bray–Curtis coefficient. Indicator species analysis (IndVal; Dufrene & Legendre, 1997) was used to identify significant indicator species that discriminated among the time periods for the species composition and biological trait data. IndVal is based on a comparison of relative abundance and relative frequencies of taxa in different *a priori* groups. Good indicator taxa are those occurring at all sites in a given group and never in any other groups (Dufrene & Legendre, 1997). The indicator value ranges from 0 to 100 and is maximized

when all individuals occur within a single group of sites. The significance of the indicator values for each taxon was tested by Monte Carlo tests with 1000 permutations. All ordinations, MRPP and indicator species analyses were performed in PC-Ord ver. 5 (McCune & Mefford, 1999).

Results

Hydrological and climatic patterns

Standardized precipitation index values ranged from -2.54 to 4.29 during the 50-year period from 1956 to 2006 in southwest Georgia (Fig. 1). Values >2 are classified as extremely wet and values below -2 as extremely dry (Guttman, 1999). Mean ($\pm$ SD) values for the 3-, 12- and 48-month SPI during the 1998–2002 drought were -0.25 ± 1.06 , -0.29 ± 1.42 and 0.55 ± 0.86 respectively. The drought prior to the study period (1998–2002) was the worst of the past 50 years and the third worst of the past 100 years, exceeded only by droughts from 1930 to 1935 and 1938 to 1944 (Barber & Stamey, 2000). The 1998–2002 drought had serious impacts on streams and rivers in the region, with the number of zero-flow days reaching a recurrence interval of 20–50 years and the Flint River displaying a record of low daily flows (Barber & Stamey, 2000).

The present study (late 2001 to 2007) occurred during a period of average precipitation, with slightly above-average SPI values for months 3 and 12 (i.e. mean $\pm$ SD, 0.13 ± 1.02 and 0.14 ± 1.00 respectively) and a slightly below-average SPI value for the month 48 (i.e. -0.33 ± 1.22). Additionally, hydrographs recorded flow throughout most of the sampling period (Fig. 2) and the number of zero-flow days decreased progressively over time in both streams, indicating a period of stream recovery. However, SPI values in 2006–2007 indicate return to a drought, an observation supported by occurrence of a substantial drought in Georgia in 2007–2008.

Environmental variables

Most environmental parameters fluctuated over time regardless of changes in precipitation or discharge (Table 2); however, some parameters changed significantly with time. Ammonia concentrations remained low throughout most of the study, but doubled in the

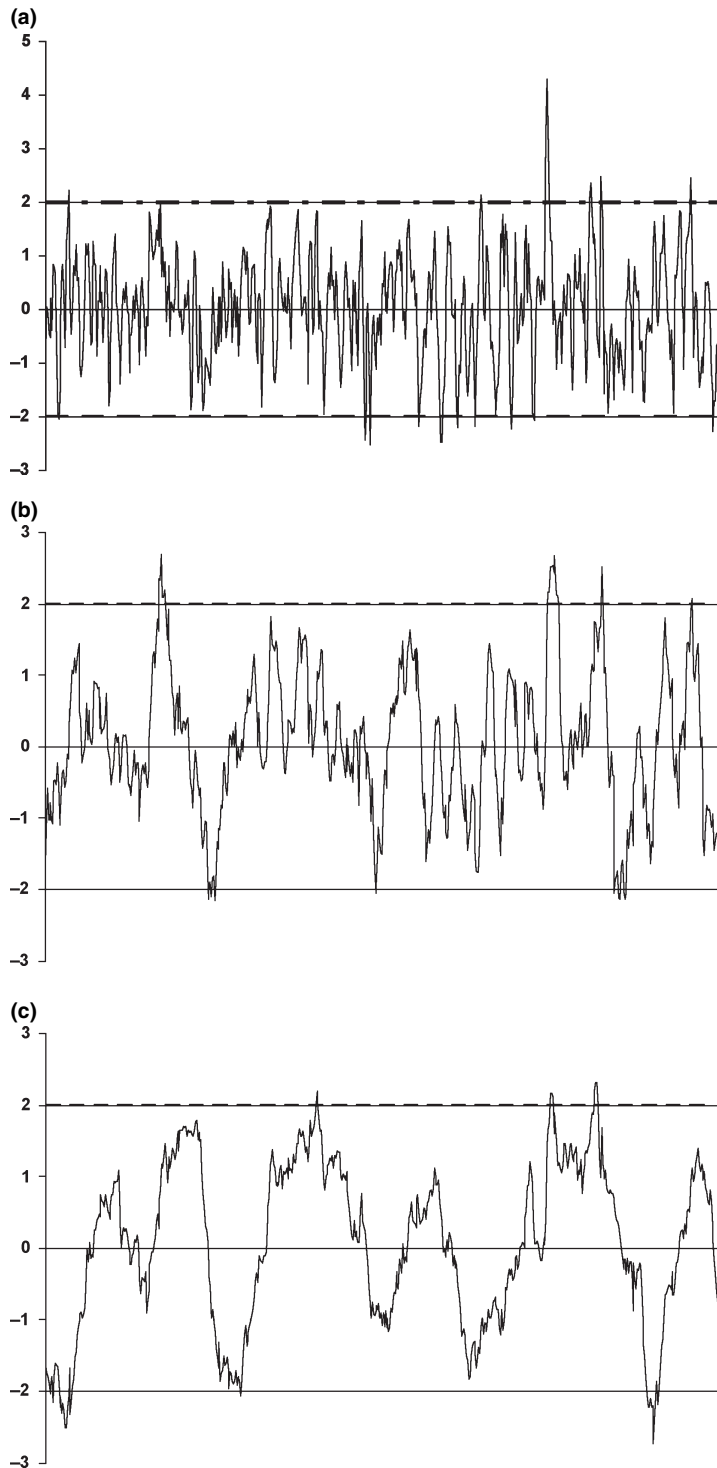


Fig. 1 Standardized precipitation index (SPI) values calculated from (a) 3-month, (b) 12-month and (c) 48-month precipitation data in southwestern Georgia, from 1956 to 2007. Values above 2 indicate an extremely wet year, while values below -2 indicate an extremely dry year.

third year in both streams ($F_{5,41} = 2.3$, $P = 0.05$). Values for pH were variable, but were highest immediately following drought, decreasing thereafter ($F_{5,41} = 4.7$, $P < 0.01$). Additionally, WF remained

more acidic than SF throughout the study. Orthophosphate decreased over time ($F_{5,41} = 5.4$, $P = 0.02$), but increased again in the 2006–2007 sampling period. In general, conductivity decreased following flow

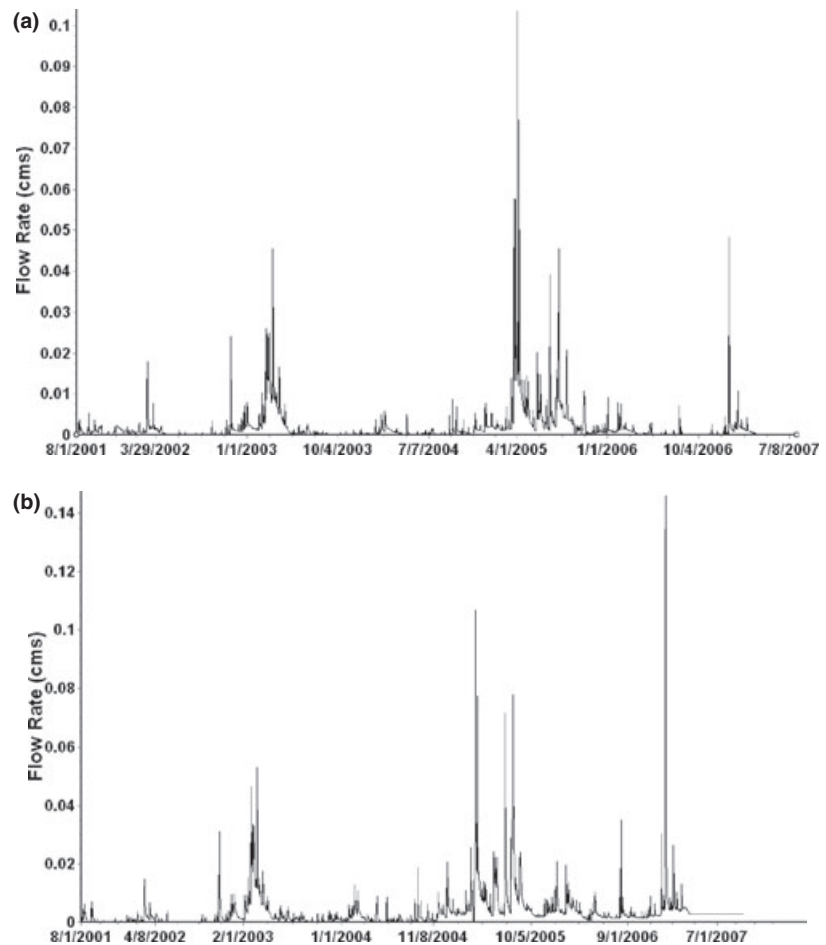


Fig. 2 Hydrograph showing mean daily discharge ($\text{m}^3 \text{s}^{-1}$) for (a) wetland and (b) seep-fed streams.

resumption ($F_{5,41} = 2.3$, $P = 0.05$) but increased again during the 2006–2007 sampling period. Temperature decreased by 4°C over the study period ($F_{5,41} = 5.1$, $P < 0.001$), ranging from 12 to 16°C . Leaf fall peaked in the first year following the drought, but was reduced by 50% the following year ($F_{5,41} = 2.6$, $P = 0.04$).

Benthic macroinvertebrates

Community succession Although the two streams differed extensively in terms of successional patterns following drought, a number of species responded similarly at both sites. A core set of species was present throughout the 6-year sampling period at both sites, including Ceratopogonidae (*Bezzia*), Chironomidae (*Parametriocnemus*, *Polypedilum*, *Tanytarsus*, *Tribeolos*, *Zavrelimyia*), Decapoda (Cambaridae), Tabanidae (*Chrysops*) and Tipulidae (*Pilaria*). Similarities in the

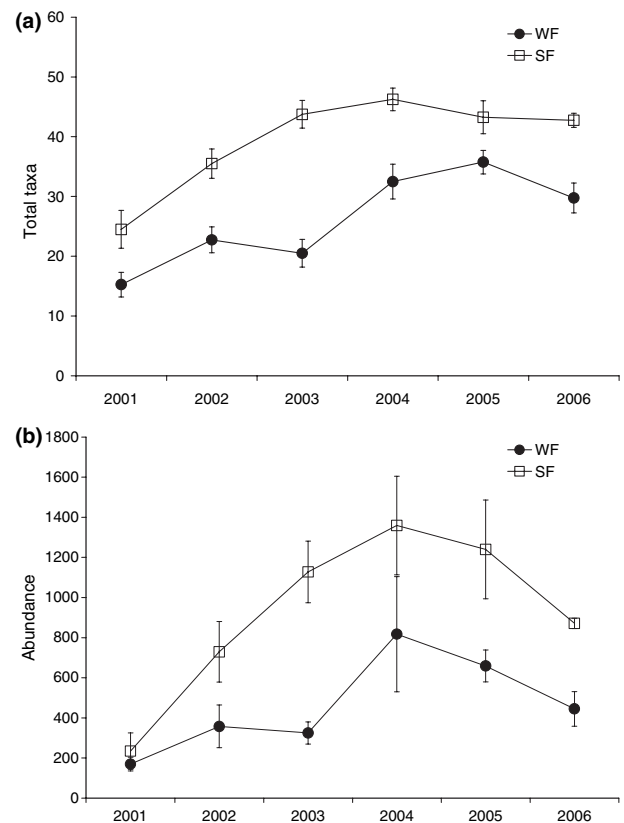
second year included Chironomidae (*Cantopelopia*, *Orthocladinae*) and Ptychopteridae (*Bittacomorpha*), while those in the fourth and fifth year of sampling included Trichoptera (*Lepidostoma*), Hemiptera (*Microvelia*) and Odonata (*Boyeria*). The sixth year was the first year that no additional taxa were found (Appendices S1 and S2).

Taxon richness increased significantly over time ($F_{4,30} = 122.73$, $P < 0.001$), primarily during the initial 3 years of the study (Fig. 3a), but was consistently lower in WF ($F_{1,30} = 56.65$, $P < 0.001$). However, taxon richness saturated with the same number of taxa occurring from the fourth to the sixth year. Temporal progression of abundance was more humped-shaped, decreasing after the fourth year. Invertebrate abundance increased similarly in both streams through time ($F_{4,30} = 8.68$, $P < 0.001$; Fig. 3b), but WF consistently had significantly fewer individuals than SF ($F_{1,30} = 19.94$, $P < 0.001$).

Table 2 Mean annual values for environmental variables for the wetland and seep-fed streams

Year	Flow (L s ⁻¹)	TSS (g L ⁻¹)	NH ₄ (µg L ⁻¹)	o-Phosphate (µg L ⁻¹)	NO ₂ /NO ₃ (µg L ⁻¹)	Total phosphorous (µg L ⁻¹)	Total nitrogen (µg L ⁻¹)	pH	Conductivity (µS cm ⁻¹)	Dissolved Oxygen (mg L ⁻¹)	Turbidity (NTU)	Temperature (°C)	Leaf fall (gm ⁻²)
WF	2001–2002	0.99	0.015	2.48	2.73	1.08	8.22	5.53	42.28	4.47	1.78	16.13	34.29
	2002–2003	2.52	0.001	6.28	1.75	0.00	3.69	4.73	30.60	7.34	0.19	12.15	12.29
	2003–2004	1.04	0.017	11.97	1.85	0.00	13.26	5.03	35.95	4.51	1.18	16.08	19.26
	2004–2005	1.76	0.003	2.88	2.49	2.34	4.90	4.87	24.90	7.62	1.23	12.00	21.26
	2005–2006	1.47	0.013	6.20	1.78	0.00	16.94	5.35	26.85	6.87	1.10	13.63	30.19
	2006–2007	2.58	0.008	3.93	3.01	0.00	9.61	5.11	33.68	8.99	0.63	13.05	24.55
SF	2001–2002	0.02	0.004	4.54	45.46	9.42	77.00	7.23	84.03	5.25	4.03	15.68	38.25
	2002–2003	2.74	0.004	0.00	27.24	7.85	285.97	5.88	94.85	7.85	2.95	12.73	16.23
	2003–2004	3.07	0.008	7.65	23.74	4.41	51.75	6.85	70.90	6.82	4.03	15.90	22.18
	2004–2005	3.39	0.003	1.68	18.86	3.00	232.05	6.61	74.98	9.39	2.98	11.73	26.06
	2005–2006	5.36	0.016	6.59	12.43	9.40	218.54	7.12	60.93	8.79	4.51	13.05	29.76
	2006–2007	3.90	0.001	4.85	25.69	2.38	203.76	7.09	82.80	9.88	2.61	12.05	25.84

WF, wetland-fed stream; SF, seep-fed stream.

Fig. 3 Temporal changes in (a) taxon richness and (b) number of macroinvertebrate individuals ($\pm$ SE) from dipnet samples of wetland (WF) and seep-fed streams (SF).

Community persistence Bray–Curtis values for taxonomic composition decreased progressively from 2001 to 2006, with increased persistence over time at both sites. However, persistence decreased during the 2006 to 2007 period, indicating a change in community structure to an earlier period (Fig. 4a). Communities were more stable in wetter than drier periods, as indicated by the negative relationship between Bray–Curtis and SPI values (Fig. 4b,c). For SF, persistence was significantly related to both local and regional hydrological and climatic indicators; however, stronger relationships existed with flow ($R^2 = 0.33$, $P < 0.001$) and the 48-month SPI ($R^2 = 0.50$, $P < 0.001$). Only long-term 48-month SPI values were related to persistence in WF, with a similar negative relationship between SPI values and persistence.

Biological traits were relatively stable over time and low Bray–Curtis values suggested that traits were more stable than taxonomic composition (Fig. 5), while those for WF were only significantly correlated

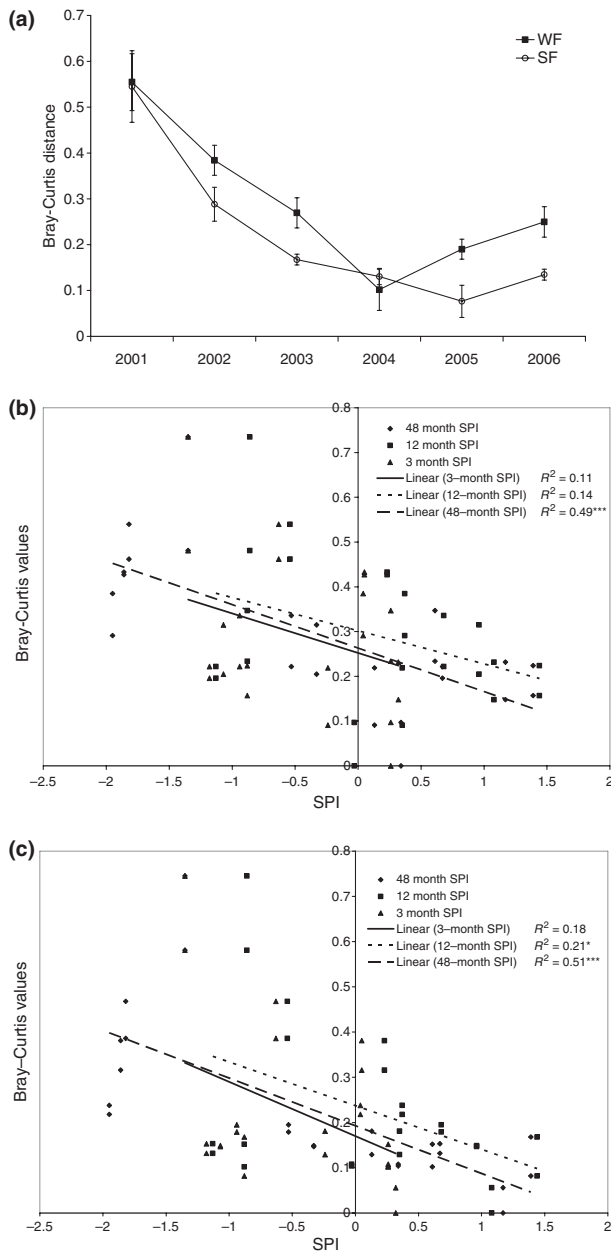


Fig. 4 Temporal changes in Bray–Curtis distances (i.e. taxonomic compositional persistence/similarity) comparing (a) successive years in wetland (WF) and seep-fed streams (SF) ($\pm$ SE) and linear regressions of 3-, 12- and 48-month SPI values versus Bray–Curtis values for (b) WF and (c) SF. Significance values are indicated by * ($P < 0.05$) and *** ($P < 0.0001$). Bray–Curtis values closer to zero indicate identical taxonomic composition.

with the intermediate 12-month SPI ($R^2 = 0.22$, $P = 0.02$). Biological traits for SF were not significantly correlated with local hydrological or regional climatic variables.

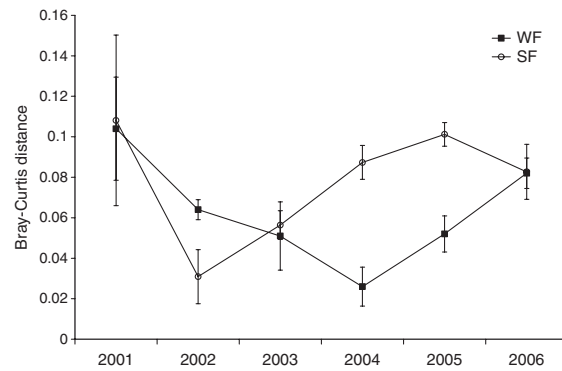


Fig. 5 Temporal changes in Bray–Curtis values based on biological traits (i.e. trait persistence) comparing successive years in wetland (WF) and seep-fed streams (SF) ($\pm$ SE). Bray–Curtis values closer to zero indicate identical trait structure.

Taxonomic composition

Wetland-fed stream Non-metric multidimensional scaling ordination (stress = 10.8, $P = 0.001$) explained 88.4% of the variance in the dataset, with 21.6%, 36% and 30.8% explained by axes 1, 2 and 3 respectively. Overall, the ordination indicated separation of species composition with time (Fig. 6a) and was supported by significant differences between all time periods (MRPP, $A = 0.46$, $P < 0.001$). Axis 1 was primarily represented by local variables including pH, dissolved oxygen and turbidity. The genera *Calopteryx*, *Chironomus*, *Chrysops*, *Eukiefferiella*, *Lepidostoma* and *Pycnopsyche* correlated most strongly with axis 1. Axis 2 was most related to both local and large-scale variables including pH, dissolved oxygen and the 12- and 48-month SPI. The genera *Agabus*, *Boyeria*, *Conchelelopia*, *Erioptera*, *Microtendipes* and *Orthocladius/Cricotopus* were most strongly related to axis 2. Axis 3 was correlated with dissolved oxygen 48-month SPI. The genera *Caecidiota*, *Microvelia* and *Smitia* were strongly correlated with axis 3 (Table 3).

No significant indicator species were found in the WF for the first 3 years following drought. Taxonomic indicators of temporal change ($P < 0.05$) included genera characteristic of the fourth year, such as *Corethrella*, *Stenochironomus*, *Polypedilum*, *Pseudolimnophila*, *Cordulegaster*, *Lepidostoma* and *Scirtidae*. Those having a maximum indicator value for the fifth year were primarily predators and included *Cryptochironomus*, *Alotanytus*, *Clinotanytus*, *Bezzia*, *Alluadomyia* and *Laevapex*. Those in the last year of the study included *Larsia* and *Erioptera*.

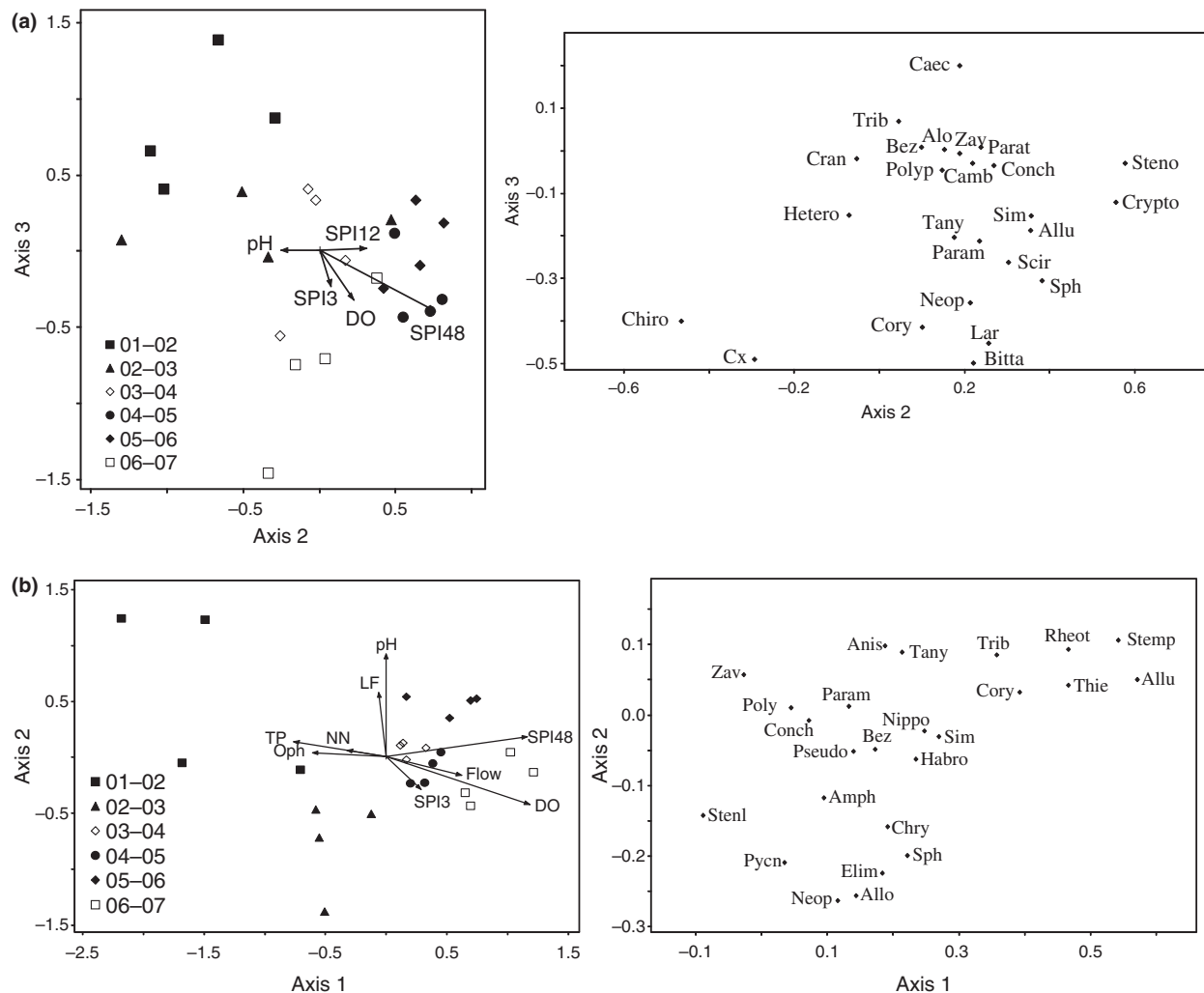


Fig. 6 Non-metric multidimensional scaling ordinations of $\log_{10}$ -abundance in site-year space and taxon space based on weighted-averaging for (a) wetland and (b) seep-fed streams. The most abundant 25 taxa are shown in the plots. Time periods are represented by different symbols; closed squares indicate the end of the drought in 2002. Arrows indicate significant vectors of maximum linear correlations between environmental variables and community composition. Species abbreviations are defined in Appendices S1 and S2.

Seep-fed stream Non-metric multidimensional scaling ordination (stress = 12.9, $P = 0.001$) explained 90.2% of the variance in the SF dataset, with 77.1% and 13.1% explained by axes 1 and 2 respectively. Overall, the ordination indicated separation of species composition with time (Fig.6b) and was supported by significant differences between all time periods (MRPP, $A = 0.49$, $P < 0.0001$). Axis 1 was primarily represented by o-phosphate, NO_2/NO_3 , dissolved oxygen, flow, 3- and 48-month SPI. The genera *Alluaudomyia*, *Bezzia*, *Corynoneura*, *Stempellinella*, *Thienemaniella* and *Zavriella* were most strongly

correlated with axis 1. Axis 2 was most related to pH, dissolved oxygen and leaf fall. The genera *Neoporus*, *Parachaetocladius*, *Sphaerium* and *Pycnopsyche* were most strongly related to axis 2 (Table 3).

Significant indicator species for the second year included *Allocaenia*, *Helichus*, *Parachaetocladius* and *Stenelmis*. The third year was mostly represented by shredders and scrapers, including *Anisocentropus*, *Cordulegaster*, *Eurylophella*, *Habrophlebiodes*, *Ophiogomphus*, *Pseudolimnophila* and *Stempellinella*. The indicators in the fourth year included Baetidae, *Psychoda*, Scirtidae, *Tribelos* and *Zavreliomyia*. The fifth year was

Table 3 Maximum correlations (*r*) of significant environmental and taxon variables in non-metric multidimensional scaling ordination space

Variables	Wetland-fed			Seep-fed	
	Axis 1	Axis 2	Axis 3	Axis 1	Axis 2
Environmental					
Orthophosphate	-0.05	-0.33	-0.10	-0.55	0.13
Nitrate–nitrite	-0.13	0.02	0.07	-0.40	0.17
pH	-0.55	-0.41	-0.03	0.03	0.68
Dissolved oxygen	0.49	0.38	-0.47	0.77	-0.46
Turbidity	-0.43	0.00	0.12	-0.06	0.04
Leaf fall	-0.34	-0.34	0.05	-0.17	0.54
Flow	0.20	0.17	0.06	0.56	-0.29
SPI3	0.37	0.25	-0.39	0.42	-0.28
SPI12	-0.20	0.41	0.14	0.18	0.09
SPI48	-0.11	0.65	-0.45	0.69	0.35
Taxon					
<i>Agabus</i>	-0.17	0.71	-0.17	–	–
<i>Alluaudomyia</i>	-0.23	0.19	-0.16	0.63	0.08
<i>Bezzia</i>	-0.10	0.28	0.02	0.66	-0.28
<i>Boyeria</i>	-0.15	0.66	-0.04	0.05	-0.09
<i>Caecidota</i>	-0.42	0.50	-0.61	-0.42	-0.02
<i>Calopteryx</i>	0.62	0.18	0.28	0.44	0.48
<i>Chironomus</i>	0.61	0.06	-0.54	–	–
<i>Chrysops</i>	0.62	0.12	-0.23	0.38	-0.47
<i>Conchepelepiia</i>	-0.22	0.64	-0.61	0.32	-0.05
<i>Corynoneura</i>	0.06	0.37	0.03	0.65	0.08
<i>Erioptera</i>	-0.20	0.78	-0.11	0.39	-0.07
<i>Eukiefferiella</i>	0.68	0.25	0.07	-0.03	-0.19
<i>Lepidostoma</i>	0.63	0.11	-0.40	0.50	-0.13
<i>Microvelia</i>	-0.10	0.32	-0.75	0.42	0.28
<i>Microtendipes</i>	0.15	0.69	-0.32	-0.13	-0.20
<i>Neoporus</i>	0.03	0.32	-0.07	0.21	-0.71
<i>Orthocladus</i>	-0.17	0.66	-0.22	-0.03	-0.19
<i>Parachaetocladus</i>	0.14	0.44	-0.38	-0.18	-0.64
<i>Pycnopsycha</i>	0.65	0.16	0.07	0.06	-0.54
<i>Smittia</i>	0.45	0.38	-0.66	–	–
<i>Sphaerium</i>	0.48	0.12	0.15	0.41	-0.56
<i>Stempelina</i>	-0.17	0.40	-0.37	0.63	0.18
<i>Thienemaniella</i>	-0.31	0.05	0.09	0.80	0.11
<i>Zavrelimyia</i>	-0.09	0.51	-0.43	-0.77	0.50

Probabilities were calculated based on 1000 random permutations. Significant correlations ($P < 0.05$) are indicated in bold.

represented by *Caenis*, *Calopteryx*, *Diplectrona*, *Microvelia*, *Nippotipula*, *Rheotanytarsus* and *Stenonema*. The last year of the study was represented by *Corynoneura*.

Biological traits

Wetland-fed stream Non-metric multidimensional scaling ordination (stress = 11.5, $P = 0.001$) explained 87.2% of the variance in the dataset, with 52.6%, 9.9% and 24.8% explained by axes 1, 2 and 3 respectively.

Temporal changes were supported by overall significant differences between time periods (MRPP, $A = 0.26$, $P < 0.0001$; Fig. 7a). However, pairwise comparisons indicated weak changes in traits over time. Axis 1 was primarily represented by specific conductance and 12-month SPI and thus related to intermediate temporal changes. Soft bodied and rheophilic sprawlers with tubular, unstreamlined shapes were positively correlated with axis 1. Sclerotized, slow flow preferring, streamlined species were negatively correlated with axis 1, primarily related to the presence of streamlined amphipods and isopods. Axis 2 was negatively correlated with pH. Burrowers were strongly correlated with axis 2. Axis 3 was most related to local, short-term variables including suspended solids, dissolved oxygen, temperature and 3-month SPI. Collector-gatherers with several generations per year were positively related to axis 3. Species with one generation a year, hard shells or cases and a predatory lifestyle were negatively related to axis 3 (Table 4).

Analysis of indicator species showed that early colonizers had sclerotized, tubular unstreamlined bodies and are abundant in the drift. Later years were characterized by species that cling to the substratum, drift rarely and are hard shelled or make a case.

Seep-fed stream Non-metric multidimensional scaling ordination (stress = 7.9, $P = 0.001$) explained 96.7% of the variance in the dataset, with 79.6% and 17.2% explained by axes 1 and 2 respectively. Overall, the ordination indicated separation of species composition with time (Fig. 7b) and was supported by significant effects of time on trait composition (MRPP, $A = 0.35$, $P < 0.0001$). Axis 1 was most related to leaf fall and the 48-month SPI. Small multivoltine species abundant in drift, with soft unstreamlined/tubular bodies were positively related to axis 1. Additionally, collector-gatherers, sprawlers and climbers were positively related to axis 1. Semivoltine, streamlined species with large hardened bodies were negatively related to axis 1. Shredders and scrapers/herbivores uncommon in drift were also negatively related to axis 1. Axis 2 was correlated with NO_3/NO_2 . Small multivoltine species lacking resistance to desiccation were positively related to axis 2. Additionally these species were rheophilic, common in the drift and classified as clingers. Univoltine, medium-sized species classified as predators and burrowers were negatively related to axis 2. Species in this group did not drift from

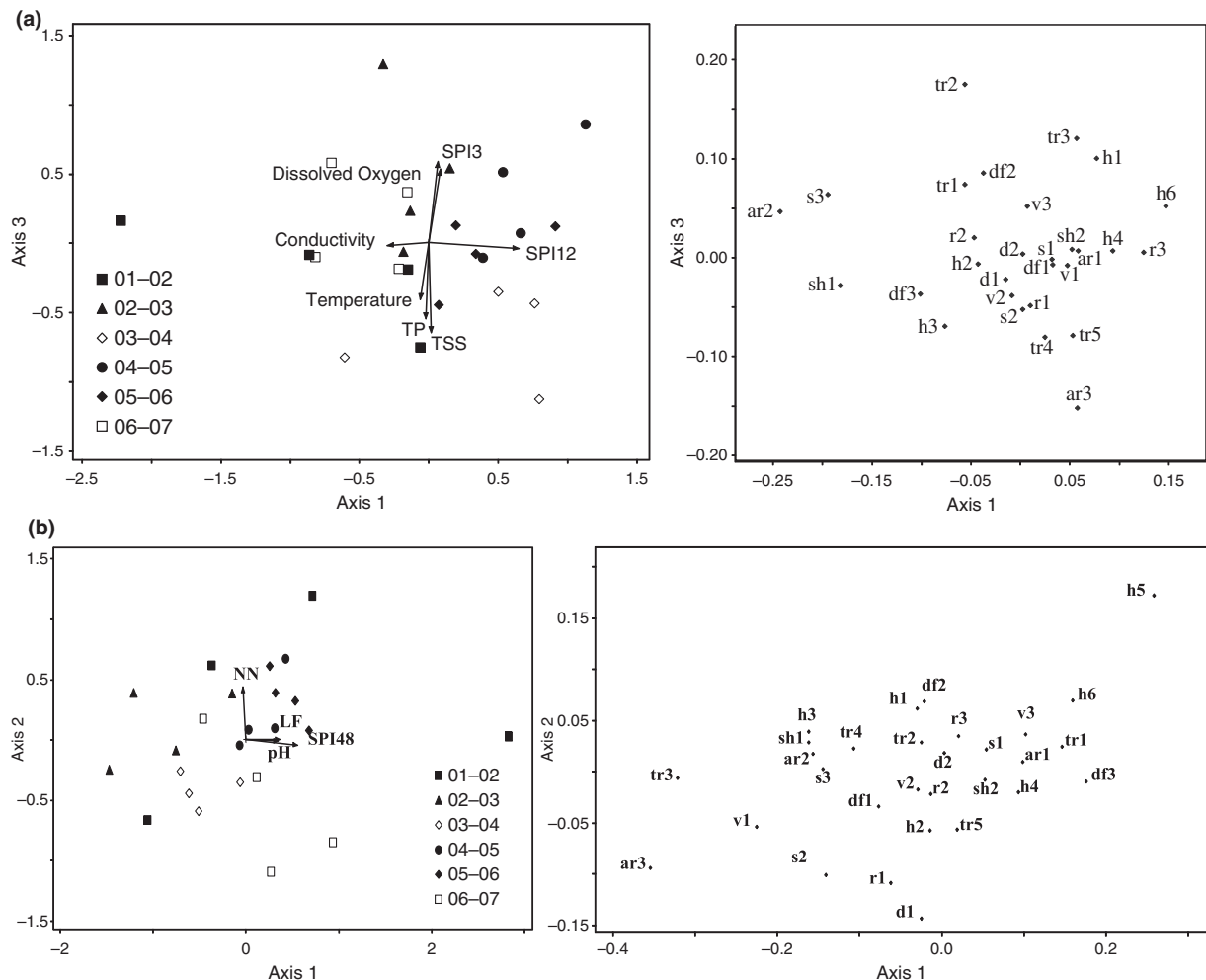


Fig. 7 Non-metric multidimensional scaling ordinations of abundance-weighted biological traits in site-year space and trait space for (a) wetland and (b) seep-fed streams. Time periods are represented by different symbols; closed squares indicate the end of the drought in 2002. Arrows indicate significant vectors of maximum linear correlations between environmental variables and trait structure. Trait abbreviations are indicated in Table 1.

unsuitable habitat but were resistant to desiccation and preferred slow flowing water (Table 4).

Indicator traits in the first 2 years included scrapers/herbivores with hard shells or cases and that crawl over the substratum. Those in the third year included genera with less than one generation per year and not resistant to desiccation. The last 2 years following the drought were represented by predators and skaters preferring fast flowing water.

Discussion

Few studies have attempted to dissect the functional and structural responses of aquatic communities to a severe, unpredictable drought (Boulton & Lake, 1992;

Wood & Petts, 1999; Wright *et al.*, 2001; Churchel & Batzer, 2006). Studies on the impacts of short-term wet and dry season cycles have provided insight into predictable climatic variation, primarily in Mediterranean and arid climates (Gasith & Resh, 1999; Acuña *et al.*, 2005; Bêche, McElravy & Resh, 2006; Bonada *et al.*, 2006). In the present study, regional precipitation indices (SPI) were good predictors of temporal changes in both taxonomic composition and biological trait structure in perennial streams. Communities became more similar over time, particularly in wet, rather than dry, years. Temporal changes in community composition and trait structure resulted in similar communities within the first 3–4 years after drought, producing persistent communities in the fourth and fifth years.

Table 4 Maximum correlations (*r*) of significant environmental and trait variables in non-metric multidimensional scaling ordination space

Variable	Wetland-fed			Seep-fed	
	Axis 1	Axis 2	Axis 3	Axis 1	Axis 2
Environmental					
TSS	0.09	0.02	-0.57	-0.09	0.08
Conductivity	-0.39	-0.19	-0.10	-0.02	-0.06
Nitrate–nitrite	-0.11	0.16	-0.53	-0.05	0.39
pH	-0.24	0.40	-0.28	-0.30	-0.36
Dissolved oxygen	0.19	0.16	0.54	0.33	0.23
Temperature	-0.17	-0.12	-0.45	-0.08	0.05
Leaf fall	-0.25	0.11	-0.02	-0.37	-0.48
SPI3	0.22	-0.03	0.44	0.11	-0.02
SPI12	0.53	0.21	-0.20	0.07	0.27
Traits					
v1	0.12	0.19	-0.03	0.81	0.37
v2	-0.10	0.35	-0.66	0.56	0.00
v3	0.07	-0.39	0.64	-0.87	-0.26
d1	-0.06	-0.31	-0.11	0.35	-0.54
d2	0.06	0.31	0.11	-0.35	0.54
df1	0.50	0.34	-0.15	0.87	0.15
df3	-0.43	-0.10	-0.22	-0.72	-0.70
ar1	0.85	0.29	0.13	-0.92	-0.63
ar2	-0.90	-0.30	0.23	0.69	0.74
ar3	0.20	0.06	-0.72	0.87	0.36
s1	0.49	-0.17	-0.05	-0.87	-0.22
s2	0.01	0.36	-0.29	0.73	-0.07
r2	-0.63	-0.02	0.35	0.36	-0.25
r3	0.60	-0.35	0.03	-0.47	0.34
h1	0.26	-0.31	0.46	-0.05	0.74
h2	-0.45	0.79	-0.10	0.39	-0.53
h4	0.60	-0.56	0.06	-0.49	-0.66
h6	0.32	-0.11	0.15	-0.67	-0.13
tr1	-0.44	-0.21	0.77	-0.80	-0.46
tr3	0.10	-0.31	0.27	0.74	0.60
tr4	0.13	0.10	-0.55	0.51	0.66
tr5	0.38	-0.01	-0.77	0.10	-0.54
sh1	-0.83	-0.36	-0.18	0.66	0.79
sh2	0.83	0.36	0.17	-0.66	-0.79

Probabilities were calculated based on 1000 random permutations. Significant correlations ($P < 0.05$) are indicated in bold.

Persistence was maintained throughout a large discharge event 4–6 months before collection of the fifth-year samples (Fig. 2). Recolonization after flood events is usually rapid, limiting effects to short-term changes in abundance and community composition (Townsend, Dolédec & Scarsbrook, 1997). Bêche *et al.* (2006) also found that invertebrate communities and trait characteristics were more stable in wet, rather than dry, season communities and postulated that droughts have more severe long-term consequences than flooding for invertebrate communities.

The representation of traits changed less with time than did taxonomic composition. This may be a

product of the high functional redundancy existing among aquatic invertebrates (Poff, 1997; Lamouroux, Dolédec & Gayraud, 2004). For example, although climatic variation can change species presence, multiple species share similar traits, meaning that traits persist even if particular species do not when conditions change.

Environmental variation

Environmental conditions were similar, but highly variable throughout the study, and did not follow changes in precipitation or flow. This may be linked to high connectivity to the floodplain and oxygenation of

the hyporheic zone; interactions typically lost during severe drying events (Boulton, 2003; Lake, 2003).

Droughts act on local stream variables by concentrating nutrients and organic matter and potentially increasing temperature (Closs & Lake, 1995; Stanley *et al.*, 1997; Matthews, 1998; Golladay & Battle, 2002; Dahm *et al.*, 2003). A decrease in ortho-phosphate following drought reflected flushing of stored nutrients during increased flow periods (Dahm *et al.*, 2003). Ammonia peaked in the third year in both streams, reflecting increased microbial activity and organic matter. Very large amounts of organic matter are typically stored in the stream channel and floodplain during drought. Initial flushes from early flow events may not have been enough to carry organic matter from the floodplain into the stream, although a large flow event in the spring prior to the third sampling period probably made a large amount of organic matter available. Baldwin *et al.* (2005) also suggested that peaks in ammonia following drying events might have originated from dead bacterial cells. As in other studies, a coupled decrease in water temperature and increase in discharge led to higher overall dissolved oxygen values (Stanley *et al.*, 1997; Matthews, 1998; Golladay & Battle, 2002). Conductivity also decreased with time, reflecting dilution of concentrated ions typically found during drought (Stanley *et al.*, 1997; Caruso, 2002) and may have been linked to greater contribution of groundwater versus surface flow found during dry periods (Caruso, 2002).

Limited precipitation and water availability in the riparian zone altered local landscape dynamics. Leaf fall within the riparian zone decreased following the drought, indicating a recovery period, as trees often drop their leaves under moisture deficit (Escudero & del Arco, 1987). Although this may provide more resources for invertebrates, leaf quality may be lower and thus limit decomposition.

Temporal variation and successional patterns in taxonomic abundance

Broad-scale measures of macroinvertebrate communities were responsive to temporal changes in environmental conditions. Overall abundance was more responsive to short-term environmental variation than taxon richness or persistence. Taxon richness was lowest immediately following drought, peaking in the 2004–2005 sampling period. Changes in taxon

richness have been linked to disturbance history in relation to short- and long-term droughts (Bêche *et al.*, 2006). As flow regimes recover, more favourable conditions exist, including increased habitat availability and heterogeneity, higher dissolved oxygen concentrations and dilution of nutrients.

A set of nine core species existed in both streams immediately following drought forming a regional species pool adapted to extreme conditions. Most either have several generations per year or a desiccation-resistant stage. Crayfish typically respond to drought by creating deeper burrows, which may also provide other species with a refuge (Boulton, 1989). The chironomid genus, *Polypedilum* makes cocoons to resist periods of drying (Hinton, 1960). The presence of coleopteran adults and hemipterans in WF immediately following drought reflect their ability to survive outside the stream and colonize rapidly via aerial dispersal (Ortega *et al.*, 1991; Wissinger, 1997).

Changes in taxonomic composition were linked to both local- and large-scale environmental variables, such as dissolved oxygen, pH, and 12- and 48-month precipitation in WF. Low pH, ranging from 3 to 6 in WF, probably excluded entire taxonomic groups as it does in many streams (Townsend, Hildrew & Francis, 1983; Rosemond *et al.*, 1992), including mayflies. However, stonefly species typically present in acidic streams (e.g. Nemouridae; Ledger & Hildrew, 2000; Dangles, 2002) were absent from WF, but present in the adjacent SF. This emphasizes the magnitude of stressors limiting invertebrate communities in WF. Dissolved oxygen is often suggested as a controlling variable for invertebrate communities in streams and is closely related to water quality (i.e. Allan, 1995). The relationship of this variable with long-term SPI values indicates the advantage of incorporating regional climatic data into bioassessment protocols.

Temporal variation in traits

Traits were more related to local abiotic variables than to indices of flow or long-term precipitation. Initially, traits may be filtered by large-scale factors, including climate and geology (Poff, 1997); thus, at the smaller scale of two adjacent catchments, traits may vary locally. Across small, physiographically homogeneous regions (e.g. catchments, ecoregions), sites are likely to be located within a single regional species pool (Zobel, 1997). Thus, as predicted by the habitat

templet model (Southwood, 1977, 1988; Townsend & Hildrew, 1994), local characteristics at the reach scale directly influence biological traits.

Traits responded predictably to changes in local environmental conditions. As pH increased and nutrients decreased, species less likely to drift became more common in the stream reaches. Initial availability of nitrogen and phosphorus allowed for early colonization by scrapers (e.g. Boulton, 1991), as algal sources within the stream accumulated on available substrata. However, this effect was not apparent in WF, reflecting the lower productivity of grazers typically associated with coloured, acidic streams (Rosemond *et al.*, 1992). Additionally, shredders became more common in the second and third year, as previously exposed patches of organic matter became submerged. In addition, species typically classified as detritivores (e.g. nemourid stoneflies) may consume algae, assuming the role of scrapers, especially in streams with lower pH (Ledger & Hildrew, 2005). In a comparison of rivers affected by drought in Italy, Fenoglio *et al.* (2007) documented an increase in collectors and a decrease in shredders and scrapers with prolonged drought duration, suggesting a predictable change in feeding guilds with drought.

After a major disturbance, mean body size of invertebrates is hypothesized to increase as communities stabilize. Accordingly, medium-sized species became more common as the community stabilized, while smaller species were more common during recovery. Small body size is often related to shorter life cycles and might therefore serve as a resilience trait. However, small body size may also allow for exploitation of refugia such as the hyporheic zone during droughts or floods (Townsend, 1989), serving as a potential source of colonizers (Dole-Olivier, Marmonier & Beffy, 1997). Thus, small body size may also be useful for resisting impacts of disturbance (e.g. Townsend *et al.*, 1997).

Organisms with longer life cycles require more stable habitat and water chemistry. Adaptations to high variability were less common by the third year of sampling, when species commonly had uni- or semi-voltine life cycles and lacked adaptations for resisting desiccation. Additionally, hardening of the exoskeleton may reduce mortality during periods of drying and floods. Sclerotized and hard-shelled organisms were more common during the first 2 years after drought, while traits favouring soft-bodied organisms

became more common with time. Although there was a general trend toward species vulnerable to desiccation, many species in the streams appeared to be adapted to some level of drying through a resistant or diapause stage.

Drought prediction

A major hurdle often encountered when assessing impacts of extreme, unpredictable events on aquatic ecosystems is the availability of data prior to the disturbance (Lake, 2000, 2003). Although long-term data (>10 years) are increasingly common in ecology, many geographical regions lack them. Long-term data indicate future instances of drought via precipitation indices, changes in persistence and trait composition. The SPI used in this study indicated a trend towards drought prior to the occurrence of such an event (Fig. 1). Values fell towards 2001 levels in 2006–2007 and a drying period was evident after the last sampling period. This observation is further supported by the occurrence of a severe drought in the same region during 2007–2008 (U.S. Drought Monitor, <http://www.drought.unl.edu/dm/archive.html>). Thus, use of long-term regional precipitation data, which are more widely available than discharge and ecological data, may provide an opportunity to study pre- and post-recovery aspects of extreme events.

Similarity of species composition decreased in both streams in 2006–2007, reflecting the oncoming drought. This change was more striking in WF than SF, probably because SF is buffered by groundwater inputs, while WF is dependent on surface water inputs from a wetland headwater. Traits were relatively stable over the study period, but became less similar between successive years in the WF from 2005–2007. Additionally, predators and species common in the drift became more abundant in 2006–2007. Predators are linked to disturbance and become more common in disturbed or degraded streams (Rawer-Jost *et al.*, 2000). This suggests a return to a disturbed condition where organisms have the ability to emigrate if environmental conditions become suboptimal, or simply die.

Biological traits have been advocated as good indicators of disturbance in aquatic ecosystems (i.e. Dolédec *et al.*, 1999). One associated issue is whether they indicate changes in disturbance regime more effectively than taxonomic structure or if they are a complementary aspect that should be examined in

disturbance ecology and biomonitoring programmes (Dolédéc & Statzner, 2008). The results are unclear, as various studies across biomes suggest they can be more, less or equally informative as taxonomic structure alone (Stevens *et al.*, 2003; Finn & Poff, 2005; Heino *et al.*, 2007; Hoeinghaus, Winemiller & Birnbaum, 2007). Although less apparent than changes in taxonomic structure, persistence of biological traits in this study indicated a disturbance during the drought period and the onset of another disturbance during the 2006–2007 sampling period. Thus, traits may be useful indicators of disturbance that can be compared across catchments. One caveat to this is that functional traits are thought to be relatively insensitive to natural variation (Charvet *et al.*, 2000; Statzner *et al.*, 2001, 2005). The present study focussed primarily on natural variation in climate and indicated changes in trait composition as a result of this variation. Thus, additional effort should be devoted to distinguishing the effects of natural and anthropogenic disturbances when devising biological monitoring programmes.

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Appendix S1. Presence or absence of genera in the wetland-fed stream (WF) by year.

Appendix S2. Presence or absence of genera in the seep-fed stream (SF) by year.

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